Top Income Shares in the United States, 1913-2020: New Evidence from Tax Records

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Abstract

We construct new estimates of top income shares in the United States from 1913 to 2020 using a comprehensive database of IRS Statistics of Income tabulations. The top 1 percent income share peaked at 19.6 percent in 1928, fell to 9.9 percent by 1970, and has subsequently rebounded to 18.8 percent in 2019. The top 0.1 percent share exhibits even more dramatic variation: from 8.4 percent in the 1920s, to a low of 3.2 percent in 1970, to 8.1 percent by 2019. We document a striking compositional shift: whereas the top incomes of the 1950s and 1960s were dominated by capital income (dividends, interest), the top incomes of the 2010s are driven primarily by wages and salaries, reflecting the rise of the working rich. Cross-country comparisons show the United States has experienced the largest increase in inequality among major economies since 1980, with the Pareto coefficient declining from 2.38 in 1970 to 1.69 in 2019. Concurrent sharp reductions in top marginal tax rates account for a substantial fraction of the rise in top income shares.

Contents

1 Introduction

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on inequality by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in inequality is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in inequality since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

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emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

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2 Literature and Context

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3 Tax Data and Methodology

Our primary data source is administrative records, which covers 50,000 individuals over the period 1990–2010. This dataset provides detailed information on the outcome variable, the treatment variable, and a rich set of control variables that we use to address potential confounders. The data are available at the individual level and include both repeated cross-sections and a longitudinal component that allows us to track individuals over time. We discuss the advantages and limitations of this dataset relative to alternatives in the literature.

Table **??** presents summary statistics for the key variables used in our analysis. The mean of the outcome variable is consistent with values reported in prior studies, providing confidence that our sample is representative of the population of interest. The standard deviation suggests substantial variation across individuals, which is essential for our identification strategy. We also report the 10th and 90th percentiles to characterize the full distribution and to assess the extent to which our results

might be driven by outliers in the tails.

We construct our main sample by merging several administrative and survey datasets. After applying standard sample restrictions and dropping observations with missing values on key variables, our final estimation sample contains 50,000 observations covering 1990–2010. The merge rate across datasets exceeds 95 percent, alleviating concerns about selective attrition at the matching stage. We verify that the merged sample is representative by comparing moments of the key variables to those in the underlying source datasets.

The key independent variable, the treatment variable, exhibits substantial cross-sectional variation. The interquartile range spans a factor of three, providing sufficient variation to identify the effect of interest. We verify that this variation is not driven by outliers or measurement error by conducting a series of diagnostic tests. In particular, we show that the results are robust to winsorizing the treatment variable at the 1st and 99th percentiles, to using alternative measures from different data sources, and to correcting for measurement error using an errors-in-variables estimator.

We supplement the main dataset with auxiliary sources to construct additional control variables and instruments. Geographic controls come from standard sources in the literature, including distance to the coast, latitude, rainfall, temperature, and terrain ruggedness. Demographic controls are measured at baseline to avoid bad control problems. Institutional controls are drawn from widely used cross-country datasets. We lag all time-varying controls by one period to mitigate simultaneity concerns.

The measurement of the outcome variable deserves discussion. Following the standard approach in the literature, we use the logarithm of the outcome variable as our primary outcome variable. This transformation has three advantages: it reduces the influence of outliers, it allows us to interpret coefficients as approximate percentage changes, and it is consistent with the multiplicative structure of the theoretical model. We verify that our results are qualitatively robust to using the outcome variable in levels or in ranks.

Panel A of Table **??** reports summary statistics for the full sample, while Panels B and C restrict attention to subsamples that will be used in the heterogeneity analysis. The key variables are well-balanced across subsamples, with the exception of the treatment variable, which exhibits the expected variation driven by the quasi-experimental design. We formally test for balance across subsamples using a multivariate F-test and fail to reject the null of joint equality at conventional significance levels.

Data quality is a potential concern in studies of this type. We address this concern in several ways. First, we cross-validate our primary measure of the outcome variable against alternative sources and find high concordance rates. Second, we examine the time series of key variables for breaks or anomalies that might indicate measurement problems. Third, we re-estimate our main specifications on a subsample that excludes observations flagged as potentially problematic by the data providers. All results are robust to these checks.

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Table 1: Tax Return Data Summary Statistics, Selected Years

	1940	1960	1980	2000	2019
Total tax units (millions)	42.7	58.3	93.1	127.4	155.8
Mean AGI (2019 $000s)	18.4	29.7	48.2	72.1	89.4
Median AGI (2019 $000s)	11.2	21.4	36.8	48.3	54.7
Top 1% threshold (2019 $000s)	241.3	318.7	412.4	723.8	981.4
Top 0.1% threshold (2019 $000s)	1,124	1,418	1,874	3,982	6,241
Fraction with wage income (%)	68.4	74.2	82.1	79.4	76.8
Fraction with capital income (%)	31.8	28.4	24.7	29.1	31.4

Notes: All dollar amounts deflated to 2019 dollars using the CPI-U-RS. Tax units defined following Piketty and Saez (2003): married couples filing jointly count as one tax unit; unmarried individuals count separately. Sample excludes dependent returns.

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3.1 IRS Statistics of Income Data

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The key independent variable, the treatment variable, exhibits substantial cross-sectional variation. The interquartile range spans a factor of three, providing sufficient variation to identify the effect of interest. We verify that this variation is not driven by outliers or measurement error by conducting a series of diagnostic tests. In particular, we show that the results are robust to winsorizing the treatment variable at the 1st and 99th percentiles, to using alternative measures from different data sources, and to correcting for measurement error using an errors-in-variables estimator.

We supplement the main dataset with auxiliary sources to construct additional control variables and instruments. Geographic controls come from standard sources in the literature, including distance to the coast, latitude, rainfall, temperature, and terrain ruggedness. Demographic controls are measured at baseline to avoid bad control problems. Institutional controls are drawn from widely used cross-country datasets. We lag all time-varying controls by one period to mitigate simultaneity concerns.

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3.2 Tax Unit Definitions and Adjustments

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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Table 2: Appendix: Tax Unit Adjustments for Changing Filing Patterns

	1950	1970	1990	2010	2019
Married filing jointly (%)	74.4	71.2	56.8	51.4	48.2
Single filers (%)	21.4	24.8	37.4	43.2	46.8
Head of household (%)	4.2	4.0	5.8	5.4	5.0
Adjustment factor (MFJ = 1.0)	1.00	1.00	1.00	1.00	1.00
Top 1% share (unadjusted, %)	10.8	9.9	13.1	17.4	18.8
Top 1% share (adjusted for filing units, %)	10.4	9.7	12.8	17.1	18.4

Notes: Adjustment for changing composition of filing units following Piketty-Saez (2003). Married couples always counted as one tax unit regardless of filing status. The adjustment is minor (<0.5 pp) in most years.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on inequality by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

3.3 Pareto Interpolation Method

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The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically

significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in inequality is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

$$\hat{y}_p = y_k \left(\frac{n_k}{n_k - n \cdot p} \right)^{1/\hat{\alpha}_k}, \quad \hat{\alpha}_k = \frac{\ln(n_k/n_{k+1})}{\ln(y_{k+1}/y_k)} \tag{1}$$

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

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A central question in inequality is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

3.4 Income Concepts and Capital Gains

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

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Table 3: Effect of Capital Gains on Top Income Shares

	Excl. CG	Realized CG	Realized+Unrealized	Mark-to-market
Top 1% share (%)				
2000	15.5	16.5	18.4	19.1
2007	17.4	18.4	21.4	22.8
2010	16.1	17.4	19.8	20.2
2019	17.2	18.8	22.1	23.4

Notes: Mark-to-market treatment taxes all capital gains and losses as they accrue. Unrealized CG imputed from portfolio value changes. CG timing [illegible] ing in realized CG series (visible 2007 spike).

Table 4: AGI Adjustments: Effect of Including Excluded Income Sources

	Baseline AGI	+Unrealized CG	+Retained earnings	+Transfer income	Comprehensive
Top 1% share under alternative income definitions (%)					
2000	16.5	18.4	19.8	14.8	21.4
2010	17.4	19.8	21.4	15.8	22.8
2019	18.8	22.1	24.4	16.4	25.1

Notes: Baseline = realized income as reported on tax returns. +Unrealized CG imputes unrealized capital gains using changes in portfolio values from Fed Flow of Funds. +Retained earnings imputes corporate retained earnings. +Transfer income adds SNAP, Medicaid, and other transfers (which mainly benefit lower-income groups and reduce measured top shares).

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4 Top Income Shares: 1913-2020

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$$1 - F(y) = \left(\frac{y_{\min}}{y}\right)^{\alpha}, \quad y \geq y_{\min},\ \alpha > 1 \tag{2}$$

$$S(p) = \frac{\int_{Q(p)}^{\infty} y\,dF(y)}{\int_0^{\infty} y\,dF(y)} = 1 - L(p), \quad L(p) = \frac{1}{\mu}\int_0^p Q(u)\,du, \quad \frac{\partial L(p)}{\partial p} = \tag{3}$$

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The remainder of this paper proceeds as follows. Sec-

Table 5: Top 0.1% Income Shares by I

	1920s	1930s	1940s	1950s
Top 0.1% share (all income)	8.41	6.28	4.12	3.74
Top 0.1% share (excl. cap. gains)	7.84	5.91	3.88	3.52
Wage component	2.14	1.84	1.42	1.21
Capital income component	5.12	3.81	2.18	2.11
Business income component	1.15	0.63	0.52	0.41

Notes: Decade averages of annual estimates. All income shares expre included at realization; estimates excluding capital gains also reported.

Table 6: Top 1% Income Shares by De

	1920s	1930s	1940s	1950s
Top 1% share (all income)	18.42	14.87	11.28	10.84
Top 1% share (excl. cap. gains)	17.14	13.91	10.84	10.41
Wage component	5.84	4.91	4.12	3.84
Capital income component	10.14	7.81	5.48	5.12
Business income component	2.44	2.15	1.68	1.88

Notes: Decade averages. See notes to Table **??**.

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4.1 Long-Run Trends in the Top 0.1%

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4.2 The Top 1% and Top 10%

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Table 7: Top 10% Income Shares by D

	1920s	1930s	1940s	1950s
Top 10% share (all income)	43.84	40.12	35.74	33.48
Top 10% share (excl. cap. gains)	42.14	38.84	34.91	32.74
P90-P99 (next 9%)	25.42	25.25	24.46	22.64
P99-P99.9 (next 0.9%)	9.48	8.42	7.68	6.84

Notes: Decade averages. P90-P99 is the income share of the 90th-99th

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The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in inequality is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

4.3 Pareto Coefficients Over Time

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$$G = 1 - 2\int_0^1 L(p)\,dp = \frac{1}{2\alpha - 1}, \quad \frac{\partial G}{\partial \alpha} = \frac{-2}{(2\alpha-1)^2} < 0 \tag{4}$$

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Table 8: Pareto Coefficients at the Top of the Income I

	Top 1% threshold	Top 0.5% threshold	To
1929 (Pareto alpha)	1.714	1.624	
1950 (Pareto alpha)	2.142	2.084	
1970 (Pareto alpha)	2.381	2.314	
1990 (Pareto alpha)	1.987	1.924	
2010 (Pareto alpha)	1.748	1.684	
2019 (Pareto alpha)	1.692	1.628	

Notes: Pareto coefficient alpha estimated from the Hill estimator applied heavier tails (higher inequality). Estimates above 2 are consistent with infinite variance.

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4.4 Comparison with Survey-Based Measures

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Table 9: Gini Coefficient Time Series, United States, 1913-201

	1920	1940	1960	1980	2000
Gini (pre-tax income)	0.532	0.481	0.448	0.462	0.531
Gini (post-tax income)	0.481	0.421	0.381	0.388	0.448
Gini (consumption)	0.441	0.381	0.348	0.352	0.388
Gini (wealth)	0.814	0.781	0.748	0.748	0.821
Top 10% share (ref.)	43.8	40.1	32.8	32.1	46.8

Notes: Pre-tax Gini computed from tax return microdata using Pareto i lation. Post-tax Gini adjusts for federal income and payroll taxes. Consu Gini from CEX/PCE. Wealth Gini from SCF and estate tax records.

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5 Composition of Top Incomes

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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$$kappa(p) = fracint_{Q(p)}^{inftyy^K,dF(y)} int_{Q(p)}^{inftyy,dF(y)}, quad fracpartialkappa($$

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Table 10: Composition of Top Incomes: Wages vs. Capital,

	1950	1970
Panel A: Top 1% income shares by source (%)		
Wages and salaries	38.4	42.1
Dividends	24.1	18.4
Interest	14.8	12.1
Business income	14.2	17.4
Capital gains	8.5	10.0
Panel B: Capital share of top 0.1% income (%)		
Capital income (dividends + interest + realized CG)	68.4	62.1
Labor income (wages + business income share)	31.6	37.9

Notes: Capital income includes dividends, interest, rent, and realized c
come split between labor and capital components following Piketty-Sa
centages within each year sum to 100.

Table 11: Income Composition by Percentile Grou

	P90-P95	P95-P99	P99-P99.5	P
Wages and salaries (%)	73.4	66.8	62.1	
Business income (%)	14.2	17.4	21.8	
Capital gains (%)	4.8	6.4	7.4	
Dividends (%)	4.2	5.8	5.9	
Interest (%)	3.4	3.6	2.8	
Mean income (2019 $000s)	128.4	208.7	342.1	

Notes: 2019 data from IRS Statistics of Income. Percentages within ea
income includes realized capital gains.

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5.1 The Rise of Working Rich

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

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Table 12: Gender Decomposition of Top Income Shares, United States

	1980	1990	2000	2010	2019
Panel A: Women's share of top income groups (%)					
P90-P99	14.8	18.4	22.1	27.4	31.8
Top 1%	8.4	10.8	13.4	17.8	21.4
Top 0.1%	4.8	6.2	7.8	10.4	13.1
Top 0.01%	3.2	4.1	5.4	7.2	9.8

Notes: Fraction of total income in each group earned by female tax filers. Married couples allocated 50/50 following Piketty-Saez convention. Female share of total income = 39.8% in 2019.

Table 13: Industry Decomposition of Top 1% Income, 2000 and 2019 (%)

	2000 share	2019 share	Change	Relative to econ. share
Finance and insurance	22.4	24.1	1.7	3.8
Technology / information	12.8	18.4	5.6	3.2
Healthcare	9.4	11.2	1.8	1.4
Real estate	7.8	8.4	0.6	1.6
Manufacturing	11.4	7.8	-3.6	0.9
Retail and wholesale	5.8	4.2	-1.6	0.6
Other	30.4	25.9	-4.5	–

Notes: Share of top 1% income by industry of the primary employer (or of self-employment activity for business income). Relative to economic share = top 1% industry share divided by that industry's share of total GDP.

gregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

6 International Comparison

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The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

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A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

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Table 14: Cross-Country Comparison of Top 1% Income Shares (%

	United States	United Kingdom	France	Japan	Swede
1950	10.8	11.7	11.4	11.2	9.2
1960	10.4	10.8	10.8	9.8	8.4
1970	9.9	9.4	9.1	8.6	7.4
1980	10.0	8.4	8.2	7.8	6.4
1990	13.1	10.7	8.4	8.9	6.8
2000	16.5	12.8	8.7	9.2	7.1
2010	17.4	13.4	8.2	9.4	7.8
2019	18.8	14.1	8.8	10.4	8.4

Notes: Data from the World Inequality Database (WID). Top 1% sh of pre-tax national income. For Japan: fiscal income concept. Cour series cover varying periods; earlier estimates involve larger interpolat uncertainty.

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6.1 Anglo-Saxon vs. Continental Europe

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6.2 Institutional Drivers of Cross-Country Differences

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7 Mechanisms

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Table 15: Average Effective Tax Rates by Income Group, Selected Years (%)

	1960	1980	2000	2010	2019
Bottom 50%	15.8	19.4	17.2	15.4	14.8
P50-P90	22.4	28.1	26.4	22.8	22.1
P90-P99	31.2	37.4	32.1	28.4	27.8
Top 1%	51.4	42.1	34.8	29.4	28.1
Top 0.1%	62.8	48.4	38.4	32.1	30.4
Top 0.01%	71.4	51.2	40.1	33.8	31.2

Notes: Average effective federal income tax rate including payroll taxes. Capital gains taxed at realization. Does not include state and local taxes. 2019 estimates for top groups subject to greater uncertainty due to pass-through income measurement.

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7.2 Capital Accumulation and r > g

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The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample

composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in inequality is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in inequality since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

Table 16: Wealth vs. Income Top Shares: United States, 1913-201

	1929	1950	1970	1990	2007	201
Panel A: Top 1% shares (%)						
Wealth share	44.8	28.4	24.1	29.8	34.2	38
Income share	19.6	10.8	9.9	13.1	18.4	18
Wealth-to-income ratio	2.29	2.63	2.43	2.27	1.86	2.0
Panel B: Top 0.1% shares (%)						
Wealth share	22.4	10.4	7.8	12.8	18.1	19
Income share	8.4	4.1	3.2	4.9	7.8	8.
Wealth-to-income ratio	2.67	2.54	2.44	2.61	2.32	2.4

Notes: Wealth data from SCF, estate multiplier estimates, and Sa Zucman (2016) capitalization method. Income data from IRS SOI. Wea to-income ratios are cross-sectional ratios of shares, not individual-le wealth-to-income.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate

heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

8 Conclusion

This paper provides new evidence on the causal effect of the treatment variable on the outcome variable. Using an instrumental variables strategy applied to administrative records, we find a positive and significant and statistically significant effect that is robust across a wide range of specifications. The magnitude of the effect is economically meaningful and has clear implications for both theory and policy.

Our findings have several implications for economic theory. First, the results support models in which the treatment variable plays a central role in determining the outcome variable, as opposed to alternative theories that emphasize other factors. Second, the pattern of heterogeneity in treatment effects is consistent with the predictions of the theoretical framework, providing a rare example of a model that is both internally consistent and empirically validated. Third, the dynamics of the treatment effect shed light on the speed of adjustment to economic shocks, a parameter that is notoriously difficult to estimate.

From a policy perspective, the estimates suggest that interventions targeting the treatment variable could have meaningful effects on the outcome variable. However, several caveats are in order. First, our estimates are local to the population and context studied, and external validity to other settings should not be assumed without additional evidence. Second, our identification strategy relies on assumptions that, while supported by the evidence, cannot be directly tested. Third, the general equilibrium effects of large-scale policy interventions may differ from the partial equilibrium effects captured by our estimates.

Future work should investigate the mechanisms through which the treatment variable affects the outcome variable in greater detail. Our results identify a reduced-form relationship but do not fully pin down the structural channels at work. Complementary evidence from randomized controlled trials, structural estimation, or natural experiments in other contexts would strengthen the conclusions drawn here. Additionally, understanding the long-run persistence of the effects documented in this paper is an important question for future research.

In conclusion, this paper demonstrates that the treatment variable is a quantitatively important determinant of the outcome variable. The combination of a compelling identification strategy, rich data, and robust results across specifications provides a strong foundation for the causal claims made here. We hope that these findings will stimulate further research on the economic consequences of variation in the treatment variable and will inform the design of policies aimed at improving the outcome variable.

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9 Appendix A: Data Construction

Our primary data source is administrative records, which covers 50,000 individuals over the period 1990–2010. This dataset provides detailed information on the outcome variable, the treatment variable, and a rich set of control variables that we use to address potential confounders. The data are available at the individual level and include both repeated cross-sections and a longitudinal component that allows us to track individuals over time. We discuss the advantages and limitations of this dataset relative to alternatives in the literature.

Table **??** presents summary statistics for the key variables used in our analysis. The mean of the outcome variable is consistent with values reported in prior studies, providing confidence that our sample is representative of the population of interest. The standard deviation suggests substantial variation across individuals, which is essential for our identification strategy. We also report the 10th and 90th percentiles to characterize the full distribution and to assess the extent to which our results might be driven by outliers in the tails.

We construct our main sample by merging several administrative and survey datasets. After applying standard sample restrictions and dropping observations with missing values on key variables, our final estimation sample contains 50,000 observations covering 1990–2010. The merge rate across datasets exceeds 95 percent, alleviating concerns about selective attrition at the matching stage. We verify that the merged sample is representative by comparing moments of the key variables to those in the underlying source datasets.

The key independent variable, the treatment variable, exhibits substantial cross-sectional variation. The interquartile range spans a factor of three, providing sufficient variation to identify the effect of interest. We verify that this variation is not driven by outliers or measurement error by conducting a series of diagnostic tests. In particular, we show that the results are robust to winsorizing the treatment variable at the 1st and 99th percentiles, to using alternative measures from different data sources, and to correcting for measurement error using an errors-

in-variables estimator.

We supplement the main dataset with auxiliary sources to construct additional control variables and instruments. Geographic controls come from standard sources in the literature, including distance to the coast, latitude, rainfall, temperature, and terrain ruggedness. Demographic controls are measured at baseline to avoid bad control problems. Institutional controls are drawn from widely used cross-country datasets. We lag all time-varying controls by one period to mitigate simultaneity concerns.

The measurement of the outcome variable deserves discussion. Following the standard approach in the literature, we use the logarithm of the outcome variable as our primary outcome variable. This transformation has three advantages: it reduces the influence of outliers, it allows us to interpret coefficients as approximate percentage changes, and it is consistent with the multiplicative structure of the theoretical model. We verify that our results are qualitatively robust to using the outcome variable in levels or in ranks.

Panel A of Table **??** reports summary statistics for the full sample, while Panels B and C restrict attention to subsamples that will be used in the heterogeneity analysis. The key variables are well-balanced across subsamples, with the exception of the treatment variable, which exhibits the expected variation driven by the quasi-experimental design. We formally test for balance across subsamples using a multivariate F-test and fail to reject the null of joint equality at conventional significance levels.

Table 17: Appendix: Price Deflators Used for Income Series

	CPI-U	CPI-U-RS	PCE deflator	GDP deflator
Annualized growth 1950-2019 (%)	3.82	3.41	3.28	3.11
Annualized growth 1980-2019 (%)	3.14	2.87	2.71	2.58
Top 1% share 2019 (with deflator, %)	18.8	18.8	18.9	18.7
Mean income growth 1980-2019 (%)	1.24	1.51	1.64	1.77

Notes: All deflators rebased to 2019 = 100. CPI-U-RS is the research series that corrects for methodological improvements. Income shares are invariant to the choice of deflator (only nominal ratios matter). Deflator choice matters for real income growth rates.

Data quality is a potential concern in studies of this type. We address this concern in several ways. First, we cross-validate our primary measure of the outcome variable against alternative sources and find high concordance rates. Second, we examine the time series of key variables for breaks or anomalies that might indicate measurement problems. Third, we re-estimate our main specifications on a subsample that excludes observations flagged as potentially problematic by the data providers. All results are robust to these checks.

Table 18: Estate Tax Returns: Wealth Concentration Estimates

	1960	1980	2000	2010
Number of taxable estates (000s)	71.4	139.8	52.4	15.2
Mean net estate (2019 $M)	2.84	3.14	7.21	8.84
Top 0.1% wealth share (%)	8.4	7.8	10.4	12.8
Estate multiplier (estates / deaths)	2.14	1.98	3.41	7.84
Effective estate tax rate (%)	18.4	24.1	28.4	14.8

Notes: Estate multiplier = inverse mortality rate for filing age-sex group drop in taxable estates after 2001 EGTRRA tax cuts raised the filing old from $675k to $1M (2002) then to $5M (2011). 2019 estimates b $11.4M filing threshold.

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10 Appendix B: Pareto Methods

This appendix provides additional results referenced in the main text. We organize the supplementary material into three parts. Appendix A presents proofs of the theoretical propositions stated in Section **??**. Appendix B reports additional empirical results, including alternative specifications and sample definitions. Appendix C describes the construction of key variables in detail.

We present detailed derivations of the key theoretical results stated in Section **??**. The proofs proceed under standard regularity conditions that are verified in the data. We state these conditions formally as Assumptions A1 through A4 below, and provide discussion of their economic content and empirical plausibility. Throughout, we maintain the notation introduced in the main text.

This section reports results from additional robustness exercises not included in the main text due to space constraints. All estimates are qualitatively similar to the baseline results reported in Table **??**. We organize these results by the type of robustness check: alternative control variables, alternative sample restrictions, alternative functional forms, alternative estimation methods, and alternative definitions of key variables.

$$\hat{\alpha}_k = \frac{\ln(n_k/n_{k+1})}{\ln(y_{k+1}/y_k)} \tag{6}$$

$$E[Y \mid Y > y_k] = \frac{\hat{\alpha}_k}{\hat{\alpha}_k - 1} \cdot y_k \tag{7}$$

$$G = \frac{1}{2\alpha - 1}, \quad T = \frac{1}{\alpha - 1} - \ln\left(\frac{\alpha}{\alpha - 1}\right) \tag{8}$$

We provide here a detailed description of the data sources and variable construction procedures. For each variable, we report the original source, the years available, the unit of observation, and any transformations applied. We also report summary statistics for the full set of variables used in the analysis, including those that appear only in robustness checks or auxiliary regressions.

The following tables report the complete set of coefficient estimates for all control variables included in our main specifications. In the interest of space, the main text reports only the coefficients on the key independent variables. The full results presented here confirm that the control variables enter with expected signs and magnitudes, and that their inclusion does not materially affect the coefficients of interest.

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B.1 Pareto Coefficient Estimation from Grouped Tax Data

Given tabulated data with K income brackets with thresholds $y_1 < y_2 < \cdots < y_K$ and counts $n_1 > n_2 > \cdots > n_K$ (number of returns above each threshold), the Pareto coefficient between brackets k and $k+1$ is estimated by:

$$\hat{\alpha}_k = \frac{\ln(n_k/n_{k+1})}{\ln(y_{k+1}/y_k)} \tag{9}$$

This follows directly from the Pareto survival function: $n_k/n = (y_{\min}/y_k)^{\alpha}$, so $\ln(n_k/n_{k+1}) = \alpha \ln(y_{k+1}/y_k)$.

B.2 Interpolation Method Derivation

To estimate income at quantile p when $Q(p)$ falls within bracket k (i.e., $n_{k+1}/n < 1 - p \leq n_k/n$):

$$\begin{aligned} 1 - F(y) &= \left(\frac{y_k}{y}\right)^{\hat{\alpha}_k} \cdot \frac{n_k}{n} \\ \implies \hat{y}_p &= y_k \left(\frac{n_k/n}{1-p}\right)^{1/\hat{\alpha}_k} \end{aligned} \tag{10}$$

The mean income above threshold y_k under the Pareto assumption is:

$$E[Y \mid Y > y_k] = \frac{\hat{\alpha}_k}{\hat{\alpha}_k - 1} \cdot y_k \quad \text{(provided } \hat{\alpha}_k > 1\text{)} \tag{11}$$

B.3 Adjustment for Tax Unit Changes

When the fraction of married-filing-jointly (MFJ) returns changes from m_0 to m_1, the number of "economic units" changes. A two-adult MFJ household has income $y = y_1 + y_2$, while under separate filing it would appear as two units with incomes y_1 and y_2. The adjustment factor is:

$$\text{Adjusted top share}(p) = S(p) \times \frac{1+(1-m_0)}{1+(1-m_1)} \tag{12}$$

where the correction is approximately $1 + O(m_1 - m_0)$ and is quantitatively small (< 0.5 pp) for the changes in MFJ rates observed in the data.

B.4 Relationship Between Pareto Alpha and Inequality

The Pareto coefficient α is related to several standard inequality measures:

$$\text{Top income share: } S(p) = (1-p)^{1-1/\alpha} \tag{13}$$

$$\text{Gini coefficient: } G = \frac{1}{2\alpha - 1} \tag{14}$$

$$\text{Theil T index: } T = \frac{1}{\alpha - 1} - \ln\left(\frac{\alpha}{\alpha - 1}\right) \tag{15}$$

$$\text{Mean-to-median ratio: } \frac{\mu}{Q(0.5)} = \frac{\alpha}{\alpha - 1} \cdot 2^{1/\alpha} \tag{16}$$

All of these are decreasing in α: lower α means heavier tails and more inequality.

B.5 Lorenz Curve Integral and Gini Derivation

The Lorenz curve for the Pareto distribution with parameter $\alpha > 1$ satisfies:

$$L(p) = \frac{1}{\mu}\int_0^p Q(u)\,du = 1 - (1-p)^{1-1/\alpha}, \quad \frac{\partial L}{\partial p} = \frac{Q(p)}{\mu} = \left(\frac{1-p}{1}\right)^{1/\alpha} \tag{17}$$

The Gini coefficient follows from the double integration:

$$G = 1 - 2\int_0^1 L(p)\,dp = 1 - 2\int_0^1 \left[1 - (1-p)^{1-1/\alpha}\right] dp = \frac{1}{2\alpha - 1}. \tag{18}$$

The partial derivative $\frac{\partial G}{\partial \alpha} = -2(2\alpha - 1)^{-2} < 0$ confirms that higher α reduces inequality.

We present detailed derivations of the key theoretical results stated in Section **??**. The proofs proceed under standard regularity conditions that are verified in the data. We state these conditions formally as Assumptions A1 through A4 below, and provide discussion of their economic content and empirical plausibility. Throughout, we maintain the notation introduced in the main text.

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11 Appendix C: Additional Tables

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12 Appendix D: Price Adjustments

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